Practical No: 1

Aim: To study the Private Key and Public Key cryptographic systems.

Objectives

1. To understand the basic concept of cryptography.
2. To study encryption and decryption processes.
3. To understand the working of symmetric key cryptography.
4. To understand the working of asymmetric key cryptography.
5. To compare private key and public key cryptographic systems.
6. To study their practical applications in information and network security.

Introduction

Cryptography is a fundamental technique used in information security to protect data from unauthorized access, modification, and disclosure. It involves transforming readable information, known as **plaintext**, into an unreadable form, known as **ciphertext**, through an encryption process.

The ciphertext can be converted back into the original plaintext using an appropriate decryption process.

Cryptographic systems are broadly classified into:

1. **Private Key Cryptography (Symmetric Cryptography)**
2. **Public Key Cryptography (Asymmetric Cryptography)**

1. Private Key Cryptography

Private Key Cryptography, also known as **Symmetric Key Cryptography**, uses the **same secret key** for both encryption and decryption.

Working

The sender encrypts the plaintext using a secret key. The resulting ciphertext is transmitted to the receiver. The receiver uses the same secret key to decrypt the ciphertext and recover the original plaintext.

Process:

Plaintext → Encryption + Secret Key → Ciphertext

Ciphertext → Decryption + Same Secret Key → Plaintext

Example

Consider that a sender wants to transmit the message:

Plaintext: HELLO

The sender encrypts the message using a secret key. The resulting ciphertext is transmitted to the receiver. The receiver uses the same secret key to decrypt the ciphertext and obtain:

Decrypted Message: HELLO

Common Symmetric Algorithms

- AES (Advanced Encryption Standard)
- DES (Data Encryption Standard)
- 3DES

AES is widely used in modern systems because it provides strong security and efficient performance.

Advantages

- High speed of encryption and decryption.
- Efficient for processing large amounts of data.
- Requires comparatively fewer computational resources.
- Suitable for real-time data communication.

Limitations

- Secure distribution of the secret key is difficult.
- Both parties must protect the shared key.
- Managing a large number of keys becomes difficult when many users communicate.

2. Public Key Cryptography

Public Key Cryptography, also known as **Asymmetric Key Cryptography**, uses a pair of mathematically related keys:

- **Public Key:** Can be openly distributed.
- **Private Key:** Must be kept confidential by the owner.

Different keys are used for encryption and decryption.

Working

If a sender wants to send confidential information to a receiver, the sender can encrypt the information using the receiver's public key. The receiver then uses the corresponding private key to decrypt the information.

Process:

Plaintext → Encryption + Receiver's Public Key → Ciphertext

Ciphertext → Decryption + Receiver's Private Key → Plaintext

The private key does not need to be transmitted to the sender.

Common Algorithms

- RSA
- ECC (Elliptic Curve Cryptography)
- Diffie-Hellman and ECDH are commonly used for key agreement rather than ordinary message encryption.

Advantages

- Eliminates the need to share a private decryption key.
- Supports secure key establishment.
- Can provide authentication through digital signatures.
- Useful for secure communication over untrusted networks.

Limitations

- Generally slower than symmetric encryption.
- Requires more computational resources.
- Key generation and management can be more complex.

3. Comparison Between Private Key and Public Key Cryptography

Parameter	Private Key Cryptography	Public Key Cryptography
Also known as	Symmetric Cryptography	Asymmetric Cryptography
Number of keys	One shared secret key	Two keys: public and private
Encryption	Uses secret key	Typically uses recipient's public key

Decryption	Uses the same secret key	Uses corresponding private key
Speed	Fast	Relatively slower
Key distribution	Difficult	Easier because public keys can be shared
Computational requirement	Lower	Higher
Suitable for	Bulk data encryption	Key establishment and authentication
Examples	AES, 3DES	RSA, ECC
Digital signatures	Not directly supported	Supported

4. Practical Application

Modern secure communication systems generally use **both symmetric and asymmetric cryptography**.

For example, in secure web communication:

1. The client connects to a secure server.
2. Public-key cryptography is used during the initial authentication and key-establishment process.
3. A symmetric session key is established.
4. The session key is then used for efficient encryption of the actual data exchanged between the client and server.

This combination provides both **security and performance**.

5. Applications

Private Key Cryptography

- File and disk encryption
- Database encryption
- VPN communication
- Secure data storage
- High-volume network communication

Public Key Cryptography

- Digital signatures
- Authentication
- Secure key establishment
- Digital certificates
- Secure web communication

Procedure

1. Study the fundamental concepts of cryptography.
2. Understand plaintext, ciphertext, encryption, and decryption.
3. Study the working principle of symmetric key cryptography.
4. Study the working principle of asymmetric key cryptography.
5. Identify commonly used cryptographic algorithms.
6. Compare symmetric and asymmetric cryptographic systems.
7. Study their practical applications in information and network security.
8. Understand how both systems can be combined to provide secure and efficient communication.

Result

The **Private Key (Symmetric)** and **Public Key (Asymmetric)** cryptographic systems were studied successfully. Their working principles, key management

methods, algorithms, advantages, limitations, and practical applications were understood.

Conclusion

Private and public key cryptographic systems were studied successfully. Symmetric cryptography provides fast data encryption, while asymmetric cryptography provides secure key exchange and authentication. Both are widely used together to ensure secure communication.

Practical No: 2

Aim: To study the classical encryption techniques: substitution and transposition.

1. **Substitution Cipher**
2. **Transposition Cipher**

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1. Substitution Cipher

Theory

In a **substitution cipher**, each character of the plaintext is replaced by another character according to a fixed rule.

A common example is the **Caesar Cipher**, where each letter is shifted by a fixed number.

Example with key = 3:

Plaintext : HELLO
Ciphertext: KHOOR

Because:

- H → K
- E → H
- L → O
- L → O
- O → R

Python Program – Caesar Cipher

```
def caesar_encrypt(text, key):
    result = ""

    for ch in text:
        if ch.isalpha():
            shift = ord('A') if ch.isupper() else ord('a')
            result += chr((ord(ch) - shift + key) % 26 + shift)
        else:
            result += ch
```

```
    return result


def caesar_decrypt(text, key):
    return caesar_encrypt(text, -key)


text = input("Enter plaintext: ")
key = int(input("Enter key: "))

encrypted = caesar_encrypt(text, key)
decrypted = caesar_decrypt(encrypted, key)

print("Encrypted text:", encrypted)
print("Decrypted text:", decrypted)
```

Output

```
Enter plaintext: HELLO WORLD
Enter key: 3
Encrypted text: KHOOR ZRUOG
Decrypted text: HELLO WORLD
```

---

2. Transposition Cipher

Theory

In a **transposition cipher**, the characters of the plaintext are **rearranged** without changing the characters themselves.

For example:

Plaintext: HELLOWORLD

The letters are arranged in rows and then read column-wise.

Python Program – Columnar Transposition

```
def encrypt(text, key):
    text = text.replace(" ", "")
    result = ""

    for col in range(key):
```

```
        for i in range(col, len(text), key):
            result += text[i]

    return result


def decrypt(cipher, key):
    rows = len(cipher) // key
    extra = len(cipher) % key

    columns = []
    index = 0

    for col in range(key):
        size = rows + (1 if col < extra else 0)
        columns.append(cipher[index:index + size])
        index += size

    result = ""

    for row in range(rows + 1):
        for col in range(key):
            if row < len(columns[col]):
                result += columns[col][row]

    return result


text = input("Enter plaintext: ")
key = int(input("Enter number of columns: "))

encrypted = encrypt(text, key)
decrypted = decrypt(encrypted, key)

print("Encrypted text:", encrypted)
print("Decrypted text:", decrypted)
```

Output

```
Enter plaintext: HELLOWORLD
Enter number of columns: 3
Encrypted text: HLOOLELWRD
Decrypted text: HELLOWORLD
```

Practical No: 3

Aim: To study the prevention mechanisms to avoid Virus and other Malware in one's PC.

Prevention Mechanisms

1. **Install Antivirus Software:** Use reliable antivirus software and keep real-time protection enabled.
2. **Keep Software Updated:** Regularly update the operating system, web browsers, applications, and security software.
3. **Enable Firewall:** Use the built-in firewall to monitor and control unauthorized network connections.
4. **Download from Trusted Sources:** Download applications only from official websites or trusted software repositories.
5. **Avoid Suspicious Links and Attachments:** Do not open unknown email attachments or click suspicious links.
6. **Use Strong Passwords:** Use unique and strong passwords for different accounts.
7. **Enable Multi-Factor Authentication:** Enable MFA wherever available for additional account protection.
8. **Avoid Pirated Software:** Cracked software and key generators may contain malicious programs.
9. **Take Regular Backups:** Maintain regular backups of important files to protect against data loss and ransomware.
10. **Use Secure Browsing:** Keep the browser updated and verify websites before entering sensitive information.
11. **Be Careful with USB Devices:** Avoid connecting unknown or untrusted USB devices to the computer.
12. **Disable Unnecessary Features:** Disable unnecessary AutoRun and Office macros when they are not required.

13. **Secure the Network:** Use strong Wi-Fi passwords and modern wireless security such as WPA2/WPA3.
14. **Monitor System Activity:** Look for unusual pop-ups, unknown applications, unexpected network activity, or significant performance changes.

Practical Procedure

1. Check whether antivirus protection is enabled.
2. Verify that the operating system and applications are updated.
3. Enable the system firewall.
4. Perform a complete malware scan.
5. Check downloaded files before opening them.
6. Remove or quarantine detected threats.
7. Enable MFA on important online accounts.
8. Create regular backups of important data.
9. Avoid suspicious websites, links, attachments, and unauthorized software.

Result

Various methods for preventing viruses and other malware were studied and the importance of safe computing practices, updated security software, firewalls, backups, and user awareness was understood.

Conclusion

Following proper security practices such as **regular updates, antivirus protection, firewall usage, safe browsing, and regular backups** significantly reduces the risk of malware infection and data loss.

Practical No: 4

Aim: Demonstrate the use of Network tools: ping, ipconfig, ifconfig, tracert, arp, netstat, whois.

Network Tools

1. ping

Purpose:
The ping command checks whether a destination host is reachable and measures the round-trip time of network packets.

Command:

ping google.com

Use:
It is useful for checking basic network connectivity and packet loss.

---

2. ipconfig

Purpose:
The ipconfig command displays the IP configuration of a Windows computer.

Command:

ipconfig

Detailed information:

ipconfig /all

Use:
It displays information such as IP address, subnet mask, default gateway, and DNS configuration.

---

3. ifconfig

Purpose:
ifconfig is traditionally used on Linux/Unix systems to display and configure network interfaces. On many modern Linux systems, ip is preferred.

Command:

ifconfig

Alternative:

ip addr

Use:
It displays network interface information such as IP addresses and interface status.

---

4. tracert

Purpose:
The tracert command on Windows identifies the network path taken by packets to reach a destination.

Command:

tracert google.com

On Linux/macOS, the equivalent command is generally:

traceroute google.com

Use:
It helps identify routing problems, delays, or problematic network hops.

---

5. arp

Purpose:
The arp command displays the ARP cache, which contains mappings between IP addresses and MAC addresses on the local network.

Command:

arp -a

Use:
It is useful for examining local IP-to-MAC address mappings.

---

6. netstat

Purpose:
The netstat command displays network connections, listening ports, and network statistics.

Command:

netstat -an

Use:
It can be used to examine active connections and listening network ports.

---

7. whois

Purpose:
The whois command provides registration information for domains or Internet resources when WHOIS/RDAP information is available.

Command:

whois example.com

Use:
It can provide information such as registrar and domain registration details.

Practical Demonstration

A basic network troubleshooting procedure can be performed as follows:

1. Use ipconfig to check the computer's IP configuration.
2. Use ping to test connectivity to a known destination.
3. Use tracert to examine the route to the destination.
4. Use arp -a to inspect local IP-to-MAC address mappings.

5. Use netstat -an to examine active connections and listening ports.
6. Use whois to obtain available domain registration information.

Result

The network tools **ping, ipconfig, ifconfig, tracert, arp, netstat, and whois** were successfully studied and their uses in network troubleshooting and analysis were understood.

Conclusion

These network tools help administrators **check connectivity, identify network problems, inspect configurations, analyze connections, and troubleshoot communication issues** efficiently.

Practical No: 5

Aim: Perform encryption and decryption of Caesar cipher. Write a script For performing these operations.

Theory

The Caesar Cipher is a simple substitution cipher in which each alphabetic character is shifted by a fixed number of positions in the alphabet.

For example, with a shift value of **3**:

- A → D
- B → E
- C → F
- X → A
- Y → B
- Z → C

During decryption, the same shift is applied in the reverse direction.

Algorithm

Encryption

1. Read the plaintext from the user.
2. Read the shift value.
3. Shift each alphabetic character by the specified value.
4. Preserve spaces, numbers, and special characters.
5. Display the encrypted message.

Decryption

1. Read the ciphertext.
2. Use the same shift value in the reverse direction.
3. Recover the original plaintext.
4. Display the decrypted message.

Python Program

```
def caesar_encrypt(text, shift):
    result = ""

    for char in text:
```

```
        if char.isupper():
            result += chr((ord(char) - ord('A') + shift) % 26 + ord('A'))
        elif char.islower():
            result += chr((ord(char) - ord('a') + shift) % 26 + ord('a'))
        else:
            result += char

    return result


def caesar_decrypt(text, shift):
    return caesar_encrypt(text, -shift)


# Input
message = input("Enter the message: ")
shift = int(input("Enter the shift value: "))

# Encryption
encrypted_message = caesar_encrypt(message, shift)
print("Encrypted Message:", encrypted_message)

# Decryption
decrypted_message = caesar_decrypt(encrypted_message, shift)
print("Decrypted Message:", decrypted_message)
```

Sample Output

```
Enter the message: Hello, World!
Enter the shift value: 3

Encrypted Message: Khoor, Zruog!
Decrypted Message: Hello, World!
```

Result

The Caesar Cipher program was successfully implemented in Python. The given plaintext was encrypted using a specified shift value and successfully decrypted back to the original message.

Conclusion

The **Caesar Cipher encryption and decryption** process was successfully implemented using Python. The experiment demonstrated the basic working principle of a substitution cipher.

Practical No: 6

Aim: To study a real-world case of a data breach in an organization and analyze its impact on Confidentiality, Integrity, and Availability (CIA Triad).

Case Study: Equifax Data Breach (2017)

Equifax experienced a major cybersecurity breach in 2017. Attackers gained unauthorized access to sensitive information of approximately **147 million people**.

1. Confidentiality

Confidentiality means protecting information from unauthorized access.

In the Equifax breach, attackers accessed sensitive personal information such as:

- Names
- Social Security numbers
- Birth dates
- Addresses
- Driver's license information
- Some credit-card information

Impact:
The unauthorized disclosure of personal information created significant privacy and identity-theft risks for affected individuals.

2. Integrity

Integrity means ensuring that data remains accurate and is not improperly modified.

In this case, the major reported impact was **unauthorized access and theft of data**, rather than widespread modification of customer records.

Impact:

- Organizations must verify that affected databases have not been altered.
- Unauthorized access can reduce confidence in the accuracy and security of organizational data.
- Investigation and validation of records may be required after a breach.

3. Availability

Availability means ensuring that authorized users can access systems and information when required.

The breach required Equifax to take systems and services offline during investigation and remediation.

Impact:

- Some online services were temporarily unavailable.
- Security investigation and recovery required significant organizational resources.
- Customers experienced difficulty accessing certain services.

CIA Triad Analysis

Security Principle	Impact of Equifax Breach
Confidentiality	Sensitive personal information was accessed and exposed by unauthorized attackers.
Integrity	Data integrity had to be investigated and validated because attackers had unauthorized access.
Availability	Some services were disrupted while the organization investigated and responded to the incident.

Causes of the Breach

Some important factors associated with the incident included:

1. A known vulnerability in Apache Struts was exploited.
2. The vulnerability was not patched in time.
3. Attackers obtained unauthorized access to Equifax systems.
4. Security monitoring and certificate-management problems contributed to the prolonged intrusion.

Preventive Measures

Organizations can reduce the risk of similar breaches by:

- Regularly applying security patches and updates.
- Using firewalls and intrusion detection systems.

- Encrypting sensitive information.
- Implementing strong access controls.
- Using multi-factor authentication.
- Regularly monitoring network activity.
- Conducting vulnerability assessments and penetration testing.
- Maintaining secure backups and an incident-response plan.

Conclusion

The Equifax data breach demonstrates how a cybersecurity incident can affect all three components of the **CIA Triad**. Confidentiality was most directly affected because sensitive personal information was exposed, while integrity and availability also required investigation, protection, and recovery measures. Therefore, organizations should implement multiple layers of security to protect sensitive data and maintain reliable services.

Practical No: 7

Aim: Study a mobile application security breach and identify threats such as data leakage, rootkits, or unauthorized access.

Case Study: Uber Mobile Application Breach (2022)

Uber experienced a major security incident in September 2022. An attacker gained access to several of the company's internal systems and posted screenshots and messages showing access to internal resources.

1. Data Leakage

Data leakage occurs when confidential or sensitive information is exposed to unauthorized people.

In the Uber incident, the attacker reportedly accessed internal systems and information, including:

- Internal source-code repositories
- Internal dashboards
- Security-related information
- Some internal company data

Impact:

- Confidential company information was exposed.
- Attackers could potentially use exposed credentials or internal information to access additional systems.
- The organization had to investigate and secure affected systems.

2. Unauthorized Access

Unauthorized access means accessing an application, account, server, or database without permission.

The incident involved unauthorized access to Uber's internal systems. Reports indicated that the attacker obtained access after compromising an employee's credentials and then accessed additional internal resources.

Impact:

- Internal systems were accessed without authorization.
- Security teams had to investigate the scope of the compromise.

- Access credentials and systems needed to be secured.

3. Rootkits

A **rootkit** is malicious software designed to maintain privileged access to a device while attempting to hide its presence.

There is **no established public evidence that a rootkit was the primary technique used in the 2022 Uber incident**. However, rootkits are an important threat to mobile applications because, if installed on a rooted/compromised device, they can potentially:

- Hide malicious processes.
- Monitor application activity.
- Access sensitive information.
- Bypass some security controls.

Therefore, rootkit detection should be considered when analyzing mobile-device security.

4. Other Mobile Application Threats

Threat	Description	Possible Impact
Data Leakage	Sensitive information is exposed	Privacy loss
Unauthorized Access	Attacker accesses systems/accounts	Account or system compromise
Rootkit	Malware hides while maintaining privileged access	Difficult detection and control
Credential Theft	User credentials are obtained by attackers	Account takeover
Malware	Malicious software installed on device	Data theft or device compromise
Insecure Storage	Sensitive data stored without adequate protection	Information disclosure

Security Measures

To protect mobile applications from these threats:

1. Use **multi-factor authentication (MFA)**.
2. Never store passwords or sensitive information in plain text.

3. Encrypt sensitive data both **at rest and in transit**.
4. Use secure APIs and authentication mechanisms.
5. Regularly update the application and operating system.
6. Monitor unusual login and access activity.
7. Use root/jailbreak detection where appropriate.
8. Apply least-privilege access controls.
9. Perform regular security testing and vulnerability assessments.
10. Maintain an incident-response plan.

Conclusion

The Uber 2022 incident demonstrates how **unauthorized access and compromised credentials** can expose an organization's internal resources. Mobile applications can face additional threats such as **data leakage, malware, rootkits, and credential theft**. Strong authentication, encryption, secure storage, access control, monitoring, and regular security testing can reduce these risks.

Practical No: 8

Aim: Illustrate different types of security attacks and map them with appropriate security services and mechanisms.

Theory

A **security attack** is an action that attempts to compromise the confidentiality, integrity, or availability of information.

Security services provide protection against attacks, while **security mechanisms** are the techniques used to implement those services.

1. Types of Security Attacks

A. Passive Attacks

In a passive attack, the attacker **observes or monitors information** without changing it.

Examples:

- **Eavesdropping:** Secretly listening to communication.
- **Traffic Analysis:** Studying communication patterns to obtain information.

B. Active Attacks

In an active attack, the attacker **modifies, disrupts, or creates information**.

Examples:

- **Masquerade:** Attacker pretends to be an authorized user.
- **Replay Attack:** Captured data is retransmitted later.
- **Message Modification:** Information is changed during transmission.
- **Denial of Service (DoS):** Making a service unavailable to legitimate users.

2. Security Services

Common security services include:

1. **Authentication** – verifies the identity of a user or system.
2. **Access Control** – prevents unauthorized use of resources.

3. **Data Confidentiality** – protects information from unauthorized disclosure.
4. **Data Integrity** – ensures that information is not improperly modified.
5. **Non-repudiation** – provides evidence that a particular party performed an action.

3. Security Mechanisms

Security mechanisms used to provide these services include:

- **Encryption**
- **Digital Signature**
- **Hash Functions**
- **Authentication protocols**
- **Access control**
- **Firewalls**
- **Anti-malware software**
- **Intrusion Detection/Prevention Systems (IDS/IPS)**

Attack → Service → Mechanism Mapping

Security Attack	Security Service	Appropriate Mechanism
Eavesdropping	Confidentiality	Encryption
Traffic Analysis	Confidentiality	Encryption, traffic-flow protection
Masquerade	Authentication	Passwords, MFA, digital certificates
Replay Attack	Authentication / Integrity	Nonces, timestamps, sequence numbers
Message Modification	Integrity	Hash/MAC, digital signature
Man-in-the-Middle (MITM)	Authentication / Confidentiality	Encryption, digital certificates, authentication
Denial of Service (DoS)	Availability	Firewall, IDS/IPS, traffic filtering
Unauthorized Access	Access Control	Authentication, authorization, access-control lists

Example

Suppose an attacker intercepts a message:

Sender → "Transfer ₹5000" → Receiver

If the attacker changes it to:

Sender → "Transfer ₹50000" → Receiver

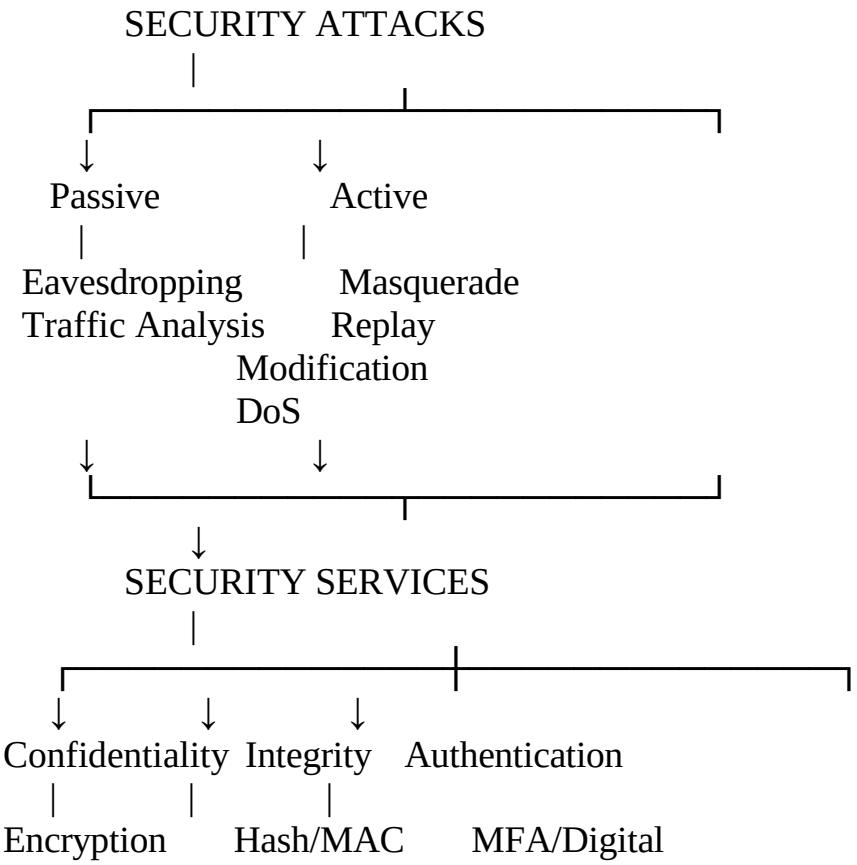
This is a **message modification attack**.

To protect it:

Attack: Message Modification
↓
Service: Data Integrity
↓
Mechanism: Hash / MAC / Digital Signature

The receiver can verify whether the message has been altered.

Simple Diagram



Certificate

↓

SECURITY MECHANISMS

Result

Thus, different types of security attacks were studied and mapped with appropriate **security services and mechanisms**. These mechanisms help protect systems against unauthorized access, information disclosure, data modification, and service disruption.